

Nebiyou Hailemariam

412-251-7954 | nebhailemariam@gmail.com | linkedin.com/in/nebhailemariam

EDUCATION

I am a graduate of Carnegie Mellon University with 5-years of experience and a strong passion for Machine Learning, Data Science, and Software Engineering. I have a solid background in analyzing data, developing models, and building and testing scalable and optimized software systems to solve complex problems. I am proficient in various deep-learning libraries like PyTorch. Additionally, I am skilled in Python, Flask, Node.js, Express.js, C#, and ASP.NET Core. My passion lies in solving complex problems that impact society.

EDUCATION

Carnegie Mellon University — CGPA 3.93

Masters in Information, Applied Machine Learning Specialization

Pittsburgh, PA

Aug. 2022 – May 2024

Addis Ababa University — CGPA 3.8

Bachelor of Science

AA, ET

Sep. 2016 – May. 2020

EXPERIENCE

Machine Learning Engineer / Software Engineer

Jul. 2021 - Jan. 2024, May 2019 - Jul. 2020

Matrix Information Technology

Remote

- Built machine learning models and developed scalable microservices, improving product recommendations, system efficiency, and backend operations across various platforms.
- Integrated Gorse, a personalized recommendation system, into an e-commerce platform to enhance product recommendations and deliver customized suggestions based on item similarity and user preferences. This led to a 20% increase in user engagement (A/B Testing).
- Used BERT-based embeddings for Amharic document classification by leveraging Hugging Face Transformers and PyTorch. Trained a multi-class neural network classifier, achieving a higher performance than previous baseline models.
- Led a team of engineers to develop and deploy scalable microservices using ASP.NET Core Web API, IdentityServer 4, and Flask on AWS EC2.
- Wrote unit and integration tests for ASP.NET Core Web API using xUnit and for Python using pytest, helping reduce bugs and ensuring robust, reliable code across services.
- Developed microservices applications for a no-code data and operations management platform using AWS (EC2, RDS, EKS, SES, SQS). Led the development of a subscription management microservice using Stripe, which simplified subscription processes and increased customer satisfaction.
- Created a microservice using Selenium that automates the generation of user dashboard screenshots and sends them via scheduled emails, improving report distribution efficiency.
- Set up CI/CD pipelines using Docker, Kubernetes, and Jenkins, optimizing deployment processes and reducing deployment time.
- Digitized a traditional financial communal saving system, by developing backend solutions using ASP.NET Core and PostgreSQL.
- Implemented key features like automated transaction processing and account management, utilizing ASP.NET Core Identity, Entity Framework, and AWS RDS for scalable and reliable database management.
- Created comprehensive unit and integration tests using xUnit, and generated realistic test data with Bogus to ensure robust system performance and reliability.
- Developed a framework to translate the SQuAD 2.0 dataset into Amharic, creating the AmaSQuAD dataset for low-resource language NLP, with applications in legal question-answering systems.
- Implemented a translation-based data generation framework for extractive question-answering (QA) and Fine-tuned XLM-R for legal question-answering in Amharic, achieving a 7% improvement in F1-score over baseline models.

Full-Stack Software Engineer

Aug. 2020 – Sep. 2021

Digital Medarbejder

Norway, Remote

- Developed a Cloud-Based School Management System for managing students, employees, and schedules, enhancing administrative efficiency and automation.
- Implemented a microservices architecture for student and employee management, and scheduling system enabling scalable and maintainable system components.

- Employed IdentityServer4 for securing microservices and ASP.NET core Identity for user Authentication.
- Leveraged Azure App Service for web hosting, Azure DevOps for CI/CD pipelines, and Azure Key Vault for managing secrets.
- Utilized various databases including Azure Cosmos DB for scalable NoSQL data storage, MongoDB for flexible document-based data management, and Azure SQL Database for relational data needs, ensuring efficient and reliable data handling.
- Wrote unit and integration tests, used Bogus for realistic test data, and documented APIs with Swagger to facilitate integration and maintenance.

Research Assistant in Machine Learning

May 2021 – Jan 2022

Empathic Computing Lab

Auckland, New Zealand

- Empathic Computing Laboratory (ECL) is an academic research laboratory directed by Prof. Mark Billinghurst at the University of South Australia in Adelaide, Australia, and the University of Auckland in Auckland, New Zealand.
- Collaborated with Ph.D. students to develop and refine methods for detecting emotions from physiological signals.
- Conducted extensive literature reviews and analyzed the performance of various machine learning and deep learning models, applying rigorous testing to ensure robustness and accuracy. Authored a 14-page paper (IUI - ACM)

PROJECTS

Machine Learning Based Rain Gauge Using Acoustic Data | Python, PyTorch, TensorFlow, Numpy, Pandas June 2020 -

- Addressed the challenges of conventional weather stations such as high setup costs, instrument fragility, and measurement errors by exploring alternative sound sources for making rainfall predictions.
- Created a Convolutional Neural Network (CNN) regression model using PyTorch and TensorFlow to estimate rainfall intensities from MFCCs extracted from acoustic recordings. Employed CNN model and achieved a Mean Absolute Percentage Error (MAPE) of 35.20% and a Mean Squared Error (MSE) of 0.66, outperforming a baseline Support Vector Regression (SVR) model with 152.55% MAPE and 1.73 MSE.

TECHNICAL SKILLS

Languages: Python, C#, C/C++, Java, Python, SQL (Postgres), JavaScript, HTML/CSS, R

Frameworks: ASP.NET Core Web API, Entity Framework, IdentityServer4, Dependency Injection, Flask, FastAPI, Django, Express.js, Node.js, Angular, React, Node.js, PyTorch, TensorFlow, Scikit-learn, Pandas, Numpy, RabbitMQ

Developer Tools: Git, Docker, Kubernetes, GCP, AWS, Azure, VS Code, Visual Studio, Postman, Jenkins, Travis CI, Eclipse, Maven, Yarn

Libraries: jQuery, Tailwind CSS, Bootstrap, Redux, Webpack

Web & Frontend Technologies: TypeScript, ReactJS, JavaScript, CSS, Bootstrap, Redux, Angular.js

Web Services: REST, SOAP, JSON, GraphQL, RESTful APIs

Cloud Services & Platforms: AWS (EC2, ECS, EKS, RDS, SES, SQS, Redshift, MQTT), Google Colab (Google Compute Engine, BigQuery, Google Cloud Storage, Google Colab)

Security: Web Security, OAuth 2.0, JWT, IdentityServer4, ASP.NET Core Identity, Passport.js

Databases & Data Storage: MySQL, PostgreSQL, MongoDB, DynamoDB, Cosmos DB, ElasticSearch, Redis, Postgres, ORM, NoSQL, SQLAlchemy, Entity Framework

Testing & Quality Assurance: xUnit, pytest, Mocha, Jest, JUnit

Development Tools & Version Control: Git, GitHub, GitLab, Bitbucket

DevOps & CI/CD: Docker, Kubernetes, Jenkins, GitHub Actions

Project Management & Collaboration: JIRA, Figma, Confluence, Trello

SDLC: Agile, Scrum

Logging & Monitoring: Prometheus, Google Analytics

Machine Learning & Deep Learning Skills: Linear Regression, Support Vector Machines, Bagging and Boosting, Neural Networks, CNNs, RNNs, LSTMs, NLP, Graph Neural Networks (GNNs), Graph Attention Networks (GATs), Hyperparameter Tuning, Transfer Learning, TensorFlow, PyTorch, Scikit-learn, XGBoost, Hugging Face Transformers, Data Preprocessing, Weights & Biases

Data Science: Data Warehousing (BigQuery, Redshift), Data Pipelines (RabbitMQ, MQTT), Data Engineering, Data Mining, Data Visualization, Data Analysis, Scikit-learn, NLP, Pandas, NumPy, PyTorch, NLTK, Tableau, PowerBI, Anaconda, PySpark, Apache Spark

Architectures: Event-Driven, Microservices, Serverless, Monolithic Architecture, Test-Driven Development (TDD), Data-Driven

Platforms: Windows, Mac OS, Unix, Linux, Jupyter, Colab, Anaconda (for Data Science)